

# Akash Debnath

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## EDUCATION

- RV College of Engineering, Bengaluru, Karnataka** **December 2021 – June 2025**  
*Bachelor of Engineering in Electronics and Communication*  
CGPA (present): 8.36 / 10
- Bishop Morrow School, Krishnagar, Nadia, West Bengal** **2013 - 2021**  
*10+2 Equivalent (Science + Maths)*  
High School  
percentage: 95%  
percentage: 96%

## PROJECTS

- RISC-V Micro-architecture Design** **2024**  
*Designed and implemented the RV32I ISA in a custom single-cycle architecture*
  - Developed a 5-stage pipelined CPU architecture
  - Dealt with various pipeline hazards and successfully made a hazard free architecture
- Direct Digital Synthesizer** (view) **2024**  
*Designed and implemented a Direct Digital Synthesis(DDS) circuit in **Tinkercad***
  - Designed using **Arduino** with an 8-bit **R-2-R DAC**
  - The circuit is capable of generating signal of **ramp, square** and **sine** waveforms
  - Achieved precise frequency generation through hardware integration
- Traffic Signal Controlling System** **2022**  
*Designed and implemented a 4-way traffic controlling system using **Verilog** in **Vivado***
  - Successfully verified the functionality of the design by generating **behavioural waveform**
- Robotic Arm Control System** (view) **2022**  
*Designed and simulated a robotic arm control system in **Tinkercad***
  - Designed using four servo motors interfaced with Arduino
  - Successfully controlled the angular positions of the **servo motors** using **serial communication**

## SKILLS

- Tools/IDEs:** LT Spice | Tinkercad | Keil | STM32 cube | Vivado | Cadence | VS code | EDA Playground | Arduino IDE
- Programming Languages:** C | C++ | System C | Embedded C
- Hardware Description Languages:** Verilog
- Assembly Languages:** ARM | MIPS | RISC-V
- Architectures:** ARM Cortex M3/M4 | Intel 8051
- Hardware Expertise:** Arduino UNO | STM32F407VG board
- Software Skills:** OOPs Concepts | Basics of Data Structures and Algorithms
- Areas of Interests:** Real Time Embedded Systems Design | Digital VLSI | Microprocessors/Microcontrollers Programming and Interfacing | Digital Systems Design | Computer Organization and Architecture | OS Fundamentals | Computer Networks
- Extra Curriculars:** Fitness Freak | Gym | Chess

## WORK EXPERIENCE

- Pantech Solutions** (Github) (view) (certificate) **November 2023 – December 2023**
  - Completed 30 days internship program and obtained certification in Embedded Systems Design
  - Learn the concepts of communication protocols such as **UART, SPI, I2C, LVDS** used in embedded systems
  - Learnt the fundamentals of **Real Time Operating Systems (RTOS)**
  - Designed and implemented a **Real Time Clock (RTC)** using **Arduino UNO** in Tinkercad
  - The RTC is capable of showing time on a **LCD** display interfaced with Arduino

## ACHIEVEMENTS

- Microsoft Azure Fundamentals AI-900** (certificate) (badge)
  - Successfully qualified the proctored examination AI-900 and obtained certification with a score of 750/1000
- NPTEL MOOC - Data Science for Engineers** (certificate)
  - Successfully qualified the proctored examination and obtained certification with a score of 81%